

Here's a complete introduction markdown file for you to download and use as documentation for running The Agentic-Researcher on ASU Sol.

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The Agentic-Researcher on ASU Sol HPC Cluster

Complete Setup and Running Guide

Overview

This guide provides step-by-step instructions for running **The Agentic-Researcher** framework on ASU's Sol HPC cluster using **Apptainer** containers with **OpenCode** as the CLI agent and **Ollama** serving local **Gemma 3** models.

Why This Setup?

Component	Purpose	Cost
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The Agentic-Researcher	Sandboxed research framework with filesystem isolation	Free (MIT License)
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OpenCode	Terminal-based coding agent that reads/writes files and executes commands	Free (MIT License)
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Ollama	Local LLM server running on ASU GPUs	Free (ASU provided)
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Gemma 3 12B	Open-source model from Google	Free (Apache 2.0)
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Total cost: \$0 - All data stays on ASU's secure infrastructure.

Prerequisites

- Active ASU Sol HPC account
- SSH access to `sol-login05.asu.edu`
- Basic familiarity with Linux command line

Part 1: Start Ollama on a Compute Node

1.1 Request a GPU Node

```
``bash
```

```
interactive -t 30 -G 1 -p htc
```

This requests:

- -t 30: 30 minutes of runtime
- -G 1: 1 GPU
- -p htc: High-throughput computing partition

1.2 Load Ollama Module

```
bash
```

```
module load ollama/0.30.3
```

1.3 Start Ollama Server

```
bash
```

```
ollama-start
```

Important: Always use ollama-start on ASU Sol. The regular ollama serve command does not properly bind to the cluster's network interface.

1.4 Get Your Node's IP Address

```
bash
```

```
hostname -i
```

Example output: 10.139.121.17

Save this IP address - you will need it for OpenCode configuration.

1.5 Pull a Model

For stability on standard GPU nodes, use the 12B model:

```
bash
```

```
ollama pull gemma3:12b
```

If you have a high-memory GPU (32GB+ VRAM), you can try:

```
bash
```

```
ollama pull gemma3:27b
```

1.6 Verify Ollama is Working

```
bash
```

```
curl http://$(hostname -i):11434/api/tags
```

Expected output shows your downloaded model(s).

Keep this terminal open - Ollama must remain running.

Part 2: Install and Configure OpenCode

Open a NEW terminal and SSH to the login node:

```
bash
```

```
ssh hwang49@sol-login05.asu.edu
```

2.1 Install OpenCode

```
bash
```

```
curl -fsSL https://opencode.ai/install | bash
```

2.2 Add to PATH

```
bash
```

```
echo 'export PATH="$HOME/.opencode/bin:$PATH"' >> ~/.bashrc
```

```
source ~/.bashrc
```

2.3 Verify Installation

```
bash
```

```
opencode --version
```

Should show version 1.17.7 or later

2.4 Configure OpenCode to Connect to Ollama

Replace 10.139.121.17 with your actual compute node IP from Part 1:

```
bash
```

```
NODE_IP="10.139.121.17" # REPLACE WITH YOUR IP
```

```
cat > ~/.config/opencode/opencode.json << EOF
```

```
{
  "\$schema": "https://opencode.ai/config.json",
  "model": "ollama/gemma3:12b",
  "provider": {
    "ollama": {
      "npm": "@ai-sdk/openai-compatible",
      "name": "Ollama (local)",
      "options": {
        "baseUrl": "http://${NODE_IP}:11434/v1"
      },
    },
    "models": {
```

```
"gemma3:12b": {  
  "name": "Gemma 3 12B"  
}  
}  
}  
}  
}  
}  
}  
EOF
```

Part 3: Run The Agentic-Researcher

3.1 Navigate to the Project

bash

```
cd ~/The-Agentic-Researcher
```

3.2 Set Environment Variables (Alternative to Config)

bash

```
export OPENCODE_MODEL="ollama/gemma3:12b"
```

```
export OPENCODE_PROVIDER="ollama"
```

3.3 Launch the Researcher

bash

```
agentic-researcher --apptainer --tool opencode --yolo
```

Flag explanations:

- --apptainer: Use Apptainer container runtime
- --tool opencode: Use OpenCode as the CLI agent
- --yolo: Auto-approve tool calls (bypass permissions)

3.4 Inside the Researcher

Once you see the /workspace prompt, initialize your research project:

text

/setup_research_plan

Follow the interactive prompts to define:

- Research goal
- Evaluation metrics
- Constraints and compute budget
- Expected outputs

The agent will create:

- report.tex - Progress tracking
- TODO.md - Task management
- CLAUDE.md or AGENTS.md - Project instructions

Quick Reference Card

Starting a Session

Terminal 1 (Compute Node):

```
bash
```

```
interactive -t 30 -G 1 -p htc
```

```
module load ollama/0.30.3
```

```
ollama-start
```

```
hostname -i # Note this IP
```

```
# Keep this terminal open
```

Terminal 2 (Login Node):

```
bash
```

```
cd ~/The-Agentic-Researcher
```

```
export OPENCODE_MODEL="ollama/gemma3:12b"
```

```
export OPENCODE_PROVIDER="ollama"
```

agentic-researcher --apptainer --tool opencode --yolo

Inside the Researcher

Command	Action
/setup_research_plan	Initialize or resume research project
/help	Show all available commands
/model	Switch AI models
/compact	Summarize conversation
/exit	Exit the sandbox

Stopping Everything

Terminal 1 (Compute Node):

bash

ollama-stop

exit

Terminal 2 (Login Node):

bash

/exit # Inside researcher, or Ctrl+C

Troubleshooting

Ollama API Connection Refused

Problem: curl: (7) Failed to connect to localhost port 11434

Solution: Ollama binds to the cluster's internal IP, not localhost. Use hostname -i to get the correct IP and test with curl http://<ip>:11434/api/tags.

OpenCode Cannot Connect to API

Problem: OpenCode shows "cannot connect to API"

Solution:

1. Verify Ollama is running: `ps aux | grep ollama`
2. Check the IP in your OpenCode config matches `hostname -i` from the compute node
3. Test connectivity: `curl http://<compute-node-ip>:11434/api/tags` from login node

Model Runs Out of Memory

Problem: signal: killed error when running model

Solution: Use a smaller model:

bash

`ollama pull gemma3:12b` *# Instead of 27b*

agentic-researcher: Command Not Found

Problem: agentic-researcher not recognized

Solution: Reinstall The Agentic-Researcher:

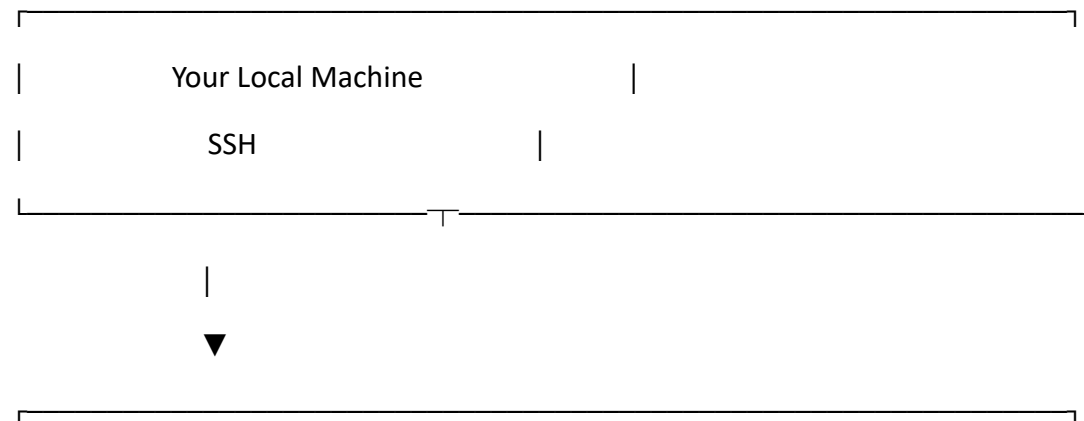
bash

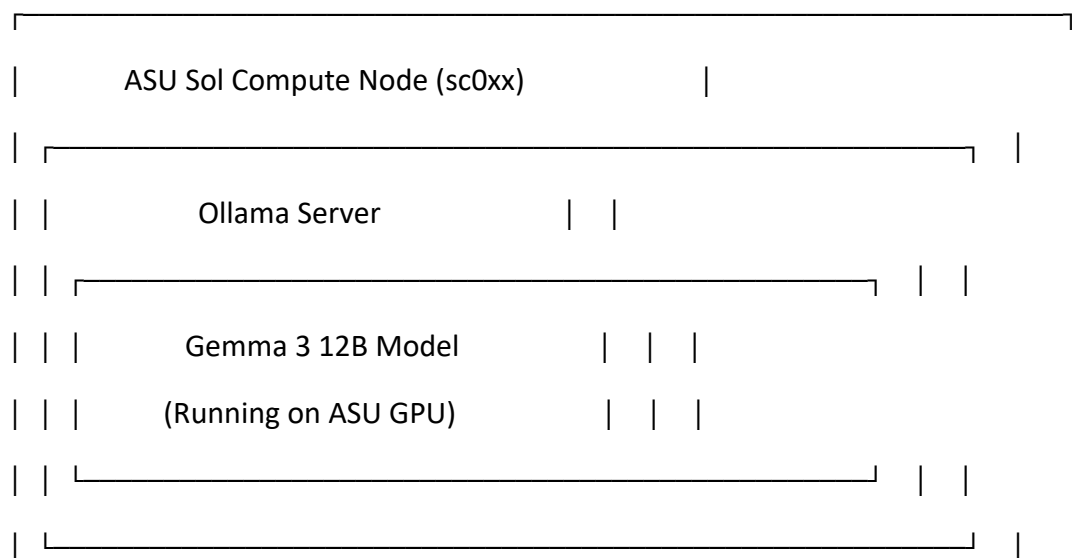
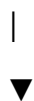
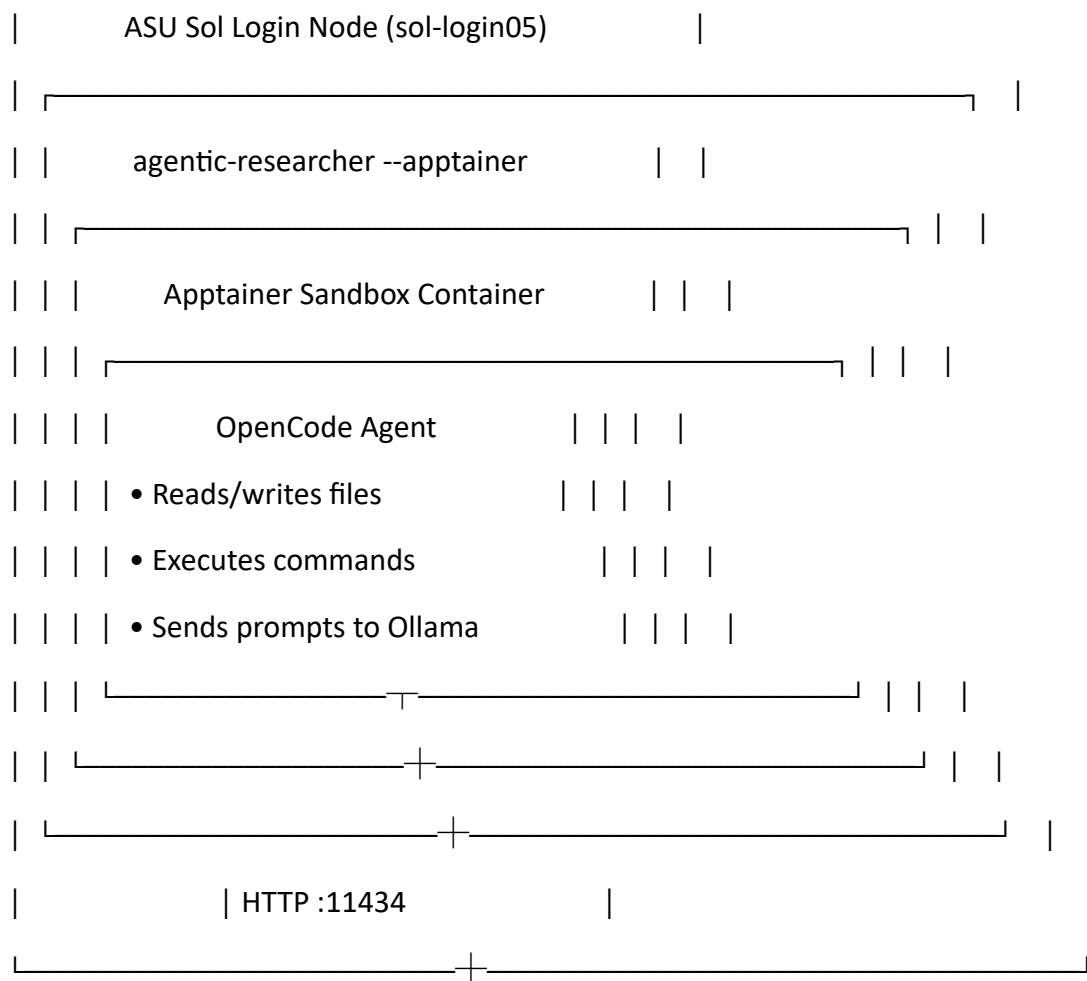
`cd ~/The-Agentic-Researcher`

`./scripts/install.sh`

Architecture Overview

text





Important Notes

Note	Explanation
Two terminals required	One for Ollama (compute node), one for researcher (login node)
Use ollama-start NOT ollama serve	ASU-specific script that properly configures networking
Save your compute node IP	OpenCode needs this to connect to Ollama
Keep Ollama terminal alive	If it dies, your agent loses the model
12B model recommended	27B may cause out-of-memory errors on standard GPU nodes
All data stays at ASU	No API calls to external services

References

- [The Agentic-Researcher GitHub](#)
- [OpenCode Documentation](#)
- [Ollama Documentation](#)
- [Google Gemma Models](#)

Support

For ASU Sol-specific issues:

- ASU Research Computing: rc.asu.edu
- Email: rc@asu.edu

For framework-specific issues:

- GitHub Issues: [ZIB-IOL/The-Agentic-Researcher/issues](https://github.com/ZIB-IOL/The-Agentic-Researcher/issues)
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Last updated: June 2026

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How to Download

Save this content to a file named `README\_ASU\_Sol.md` on your local machine:

```
```bash
Option 1: Copy the content above and save manually
Option 2: If you're on the ASU cluster, run:
cat > ~/Agentic_Researcher_ASU_Guide.md << 'EOF'
[PASTE THE ENTIRE CONTENT ABOVE HERE]
EOF
```

Or you can:

1. Select all the text in the code block above
2. Copy it (Ctrl+C or Cmd+C)
3. Open a text editor (Notepad, VS Code, etc.)
4. Paste and save as Agentic\_Researcher\_ASU\_Guide.md

The file includes all the steps from loading modules to running the researcher, plus troubleshooting and architecture diagrams.